Data Acquisition and Survey Methods

Group 10

Contents

Exercise habits and their impact on students' focus

Group 10 investigates whether students' physical exercise habits are related to their reported level of mental brain fog during lectures. The original proposal referred to exercise frequency and academic confidence. In the final survey data, the available Group 10 variables are `exercise minutes`, `preferred exercise type` and `brain fog`. Therefore, the analysis below focuses on these measured variables.

Table of contents

1. Load data
2. Survey questions and variables
3. Research questions and hypotheses
4. Exploratory data analysis
5. Descriptive inference
6. Analytic inference
7. Functions used
8. Conclusion

Load data

```
library(tidyverse)
library(kableExtra)
```

```
# Full dataset
total_survey <- read_csv("Survey_results_2026.csv", show_col_types = FALSE)

# Only Group 10 data
survey <- total_survey %>%
  select(
    starts_with("DQ"),
    starts_with("G10")
  )

# Remove strings
names(survey) <- names(survey) %>%
  str_remove("^DQ\\s*") %>%
  str_remove("^G10\\s*")

# Rename columns
survey <- survey %>%
  rename(
    age = 1,
    gender = 2,
    academic_program = 3,
    exercise_minutes = 4,
    exercise_type = 5,
    brain_fog = 6
  )

# Factorize and create new variables
survey <- survey %>%
  mutate(
    age = factor(age, levels = c("18-20", "21-23", "23-25", "25-30", "30-35", "over 35"), ordered = TRU
    exercise_type = as.factor(exercise_type),
    gender = as.factor(gender),
    academic_program = as.factor(academic_program),
    intensity_level = case_when(
      exercise_type %in% c("Weightlifting", "Cardio", "Team Sports") ~ "High-intensity",
      exercise_type == "Yoga/Flexibility" ~ "Low-intensity",
      TRUE ~ NA_character_
    ),
    intensity_level = factor(intensity_level, levels = c("Low-intensity", "High-intensity"))
  )
```

Survey questions and variables

The analysis is based on the mandatory demographic questions and the three Group 10 survey questions.

Variable	Survey question	Type
`age`	How old are you?	Quantitative
`gender`	What is your gender?	Categorical
`academic_program`	What is your academic program?	Categorical
`exercise_minutes`	How many minutes of physical exercise did you complete in the last 7 days?	Quantitative
`exercise_type`	What is your preferred type of exercise?	Categorical
`brain_fog`	Rate your average level of mental brain fog during lectures on a scale of 1 = none to 5 = severe.	Ordinal / quantitative scale

Research questions and hypotheses

RQ1: Do students who complete more minutes of physical exercise report lower levels of brain fog during lectures?
Hypothesis: Exercise minutes and brain fog are negatively associated.

RQ2: Does high-intensity training (e.g., gym, running) correlate with higher academic confidence compared to low-intensity (e.g., walking)?
Hypothesis: Students preferring high-intensity training report lower brain fog (as a proxy for higher academic confidence) than those preferring low-intensity training.

RQ3: Is age negatively associated with the number of minutes spent on physical exercise per week?
Hypothesis: Older students report fewer exercise minutes.

```r
rq1_data <- survey %>%
  filter(!is.na(exercise_minutes), !is.na(brain_fog))

rq2_data <- survey %>%
  filter(!is.na(intensity_level), !is.na(brain_fog))

rq3_data <- survey %>%
  filter(!is.na(age), !is.na(exercise_minutes))

rq1_test <- cor.test(
  rq1_data$exercise_minutes,
  rq1_data$brain_fog,
  method = "spearman",
  alternative = "less",
  exact = FALSE
)

rq2_test <- wilcox.test(brain_fog ~ intensity_level, data = rq2_data, exact = FALSE)

rq3_test <- kruskal.test(exercise_minutes ~ age, data = rq3_data)

p_text <- function(p) {
  ifelse(p < 0.001, "< 0.001", sprintf("%.3f", p))
}

brain_fog_cols <- c(
```

```r
  "1" = "#1a9850",
  "2" = "#91cf60",
  "3" = "#FFDB58",
  "4" = "#fc8d59",
  "5" = "#d73027"
)

rq2_summary <- rq2_data %>%
  group_by(intensity_level) %>%
  summarise(
    n = n(),
    mean_brain_fog = mean(brain_fog, na.rm = TRUE),
    median_brain_fog = median(brain_fog, na.rm = TRUE),
    .groups = "drop"
  )

lowest_fog_type <- rq2_summary$intensity_level[which.min(rq2_summary$mean_brain_fog)]
highest_fog_type <- rq2_summary$intensity_level[which.max(rq2_summary$mean_brain_fog)]
```

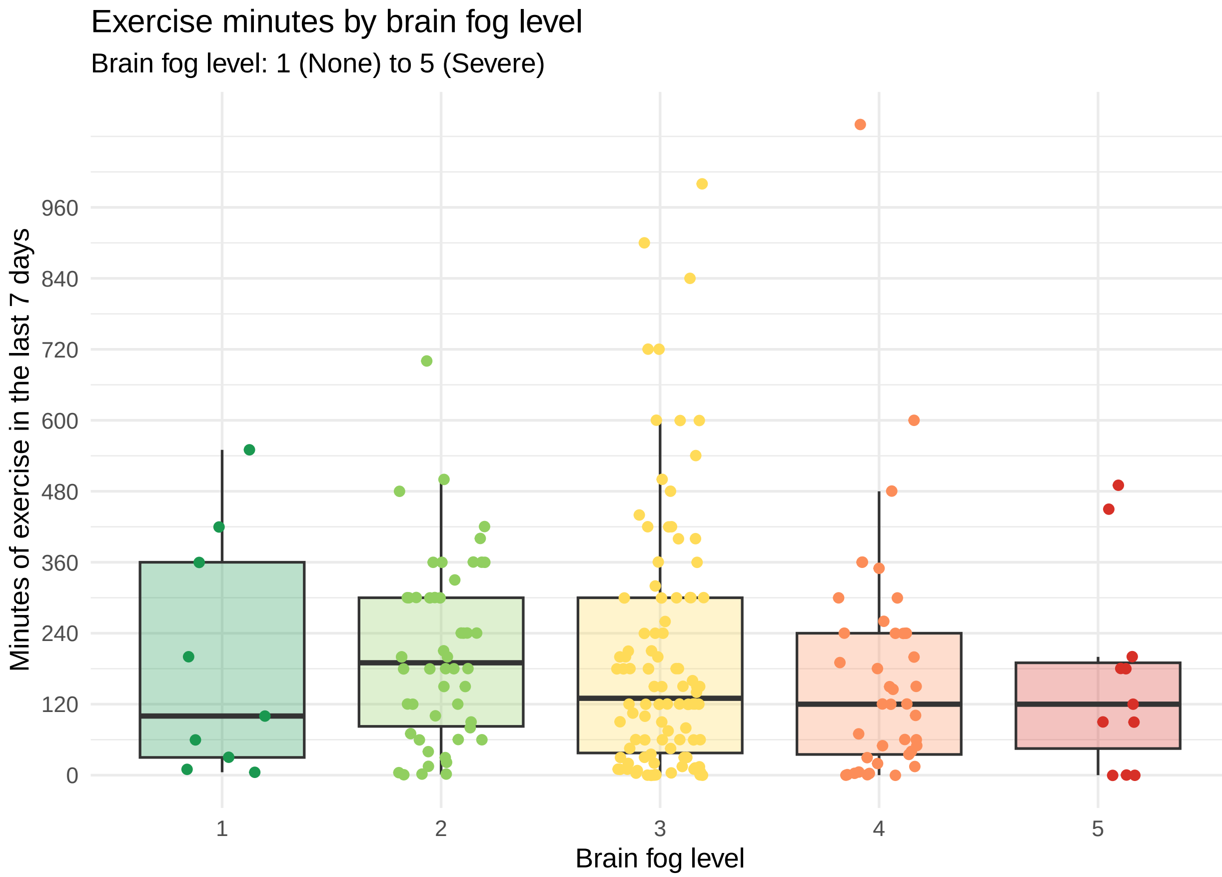
Exploratory data analysis

The exploratory analysis uses one visualization for each research question. The plots include both quantitative and categorical variables and help identify patterns before formal hypothesis testing.

RQ1: Exercise minutes and brain fog

We built a data visualization to analyze how weekly exercise minutes vary across different levels of brain fog. We converted the `brain_fog` variable into a discrete factor so R would treat it as a categorical x-axis with five distinct groups.

```r
ggplot(survey, aes(x = factor(brain_fog), y = exercise_minutes, fill = factor(brain_fog))) +
  geom_boxplot(alpha = 0.3, outlier.shape = NA) +
  geom_jitter(aes(color = factor(brain_fog)), width = 0.2, alpha = 1, size = 1.5) +
  scale_y_continuous(breaks = c(0,120,240,360,480,600,720,840,960)) +
  scale_fill_manual(values = brain_fog_cols, name = "Brain fog level") +
  scale_color_manual(values = brain_fog_cols, guide = "none") +
  labs(
    title = "Exercise minutes by brain fog level",
    subtitle = "Brain fog level: 1 (None) to 5 (Severe)",
    x = "Brain fog level",
    y = "Minutes of exercise in the last 7 days"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```



Finding:

Our initial hypothesis stated that exercise minutes and brain fog are negatively associated, meaning that higher levels of physical activity would correspond to lower levels of cognitive fatigue.

The empirical data, however, reveals a more nuanced, non linear relationship rather than a strict, steady downward trend. The central tendencies, indicated by the median lines, actually peak at a mild brain fog level of 2 (around 200 minutes) and show only a gradual decline through levels 3 and 4, while students reporting zero brain fog at level 1 display a surprisingly low median of approximately 100 minutes. Furthermore, the high variability and vertical spread within levels 1 through 3 including several highly active outliers in level 3 exercising up to 1,000 minutes demonstrate that moderate brain fog is common across a wide range of activity levels.

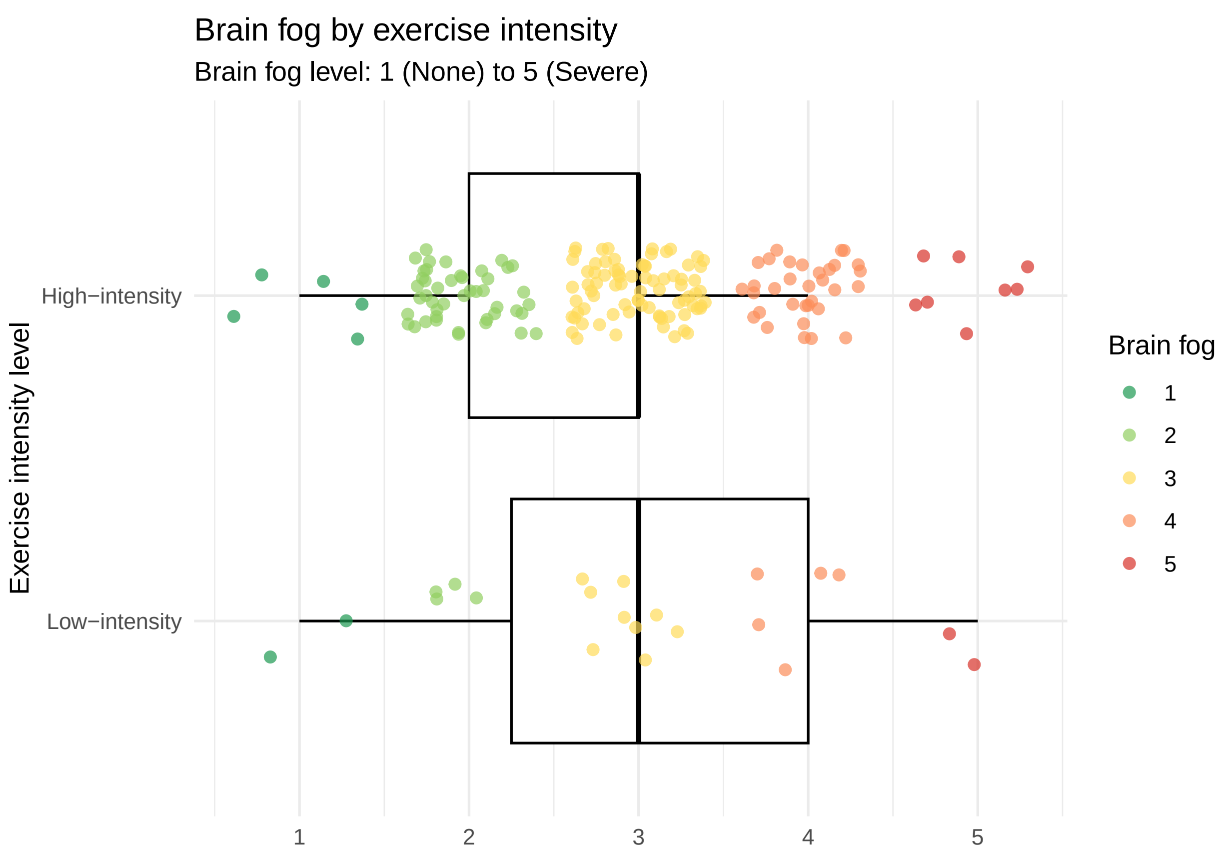
Nevertheless, our hypothesis holds true at the ultimate extreme of the spectrum. The distribution for level 5 is highly compressed and strictly bounded, showing that students experiencing severe brain fog almost never exceed 200 minutes of weekly exercise, while those who exercise intensely (beyond 400 minutes) rarely report severe cognitive symptoms.

The hypothesis fails as a linear rule because high exercise does not automatically guarantee a clear mind.

RQ2: Brain fog by exercise intensity

A boxplot is appropriate because the explanatory variable is categorical and the outcome variable is measured on a 1-5 scale. We grouped the exercise types into two intensity levels: - High-intensity (Cardio, Weightlifting, Team Sports) and - Low-intensity (Yoga/Flexibility).

```r
ggplot(survey[!is.na(survey$intensity_level), ], aes(
  x = intensity_level,
  y = brain_fog,
  color = factor(brain_fog)
)) +
  geom_boxplot(color = "black", outlier.shape = NA) +
  geom_jitter(width = 0.15, alpha = 0.7, size = 1.8) +
  coord_flip() +
  scale_y_continuous(breaks = 1:5) +
  scale_color_manual(
    values = brain_fog_cols,
    name = "Brain fog"
  ) +
  labs(
    title = "Brain fog by exercise intensity",
    subtitle = "Brain fog level: 1 (None) to 5 (Severe)",
    x = "Exercise intensity level",
    y = ""
  ) +
  theme_minimal()
```

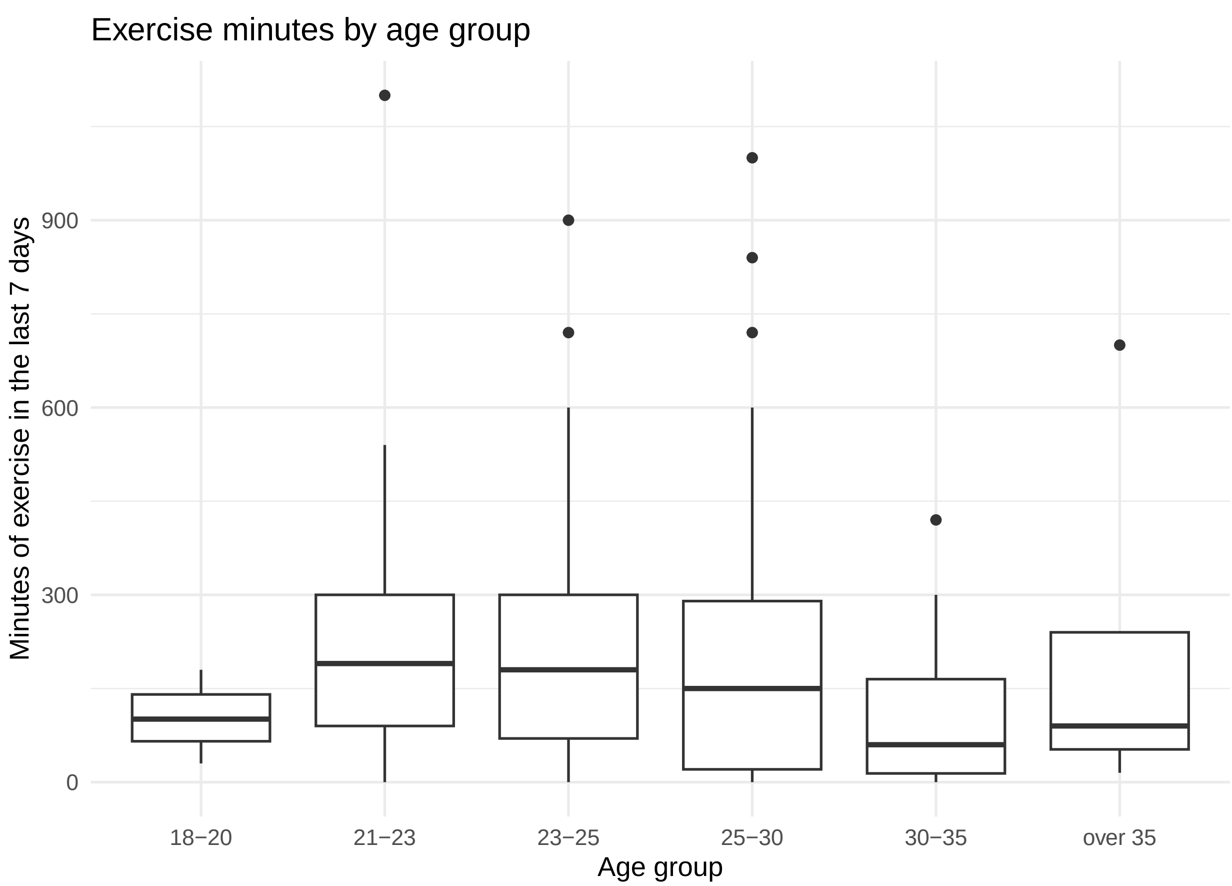


Finding: The visualization reveals that the median brain fog level is identical (level 3) for both high-intensity and low-intensity exercise groups. However, the high-intensity group has a tighter and lower interquartile range (spanning levels 2 to 3) compared to the low-intensity group (which spans levels 2 to 4). This indicates that while the medians are the same, high-intensity exercise is descriptively associated with a slightly lower overall distribution of brain fog.

RQ3: Age and exercise minutes

A scatterplot is used because both age and exercise minutes are quantitative variables. A trend line is added to show the general direction of the relationship.

```
ggplot(survey, aes(x = age, y = exercise_minutes)) +
  geom_boxplot() +
  labs(
    title = "Exercise minutes by age group",
    x = "Age group",
    y = "Minutes of exercise in the last 7 days"
  ) +
  theme_minimal()
```



Finding: The plot is used to assess whether older students tend to report fewer exercise minutes. The formal Spearman test below evaluates this negative-association hypothesis.

Descriptive inference

The following table summarizes the quantitative variables using sample size, mean, standard deviation, median, minimum, and maximum. The median is useful because exercise minutes may be right-skewed if some students report very high exercise times.

```r
# One combined descriptive summary table

numeric_vars <- survey %>%
  select(exercise_minutes, brain_fog)

numeric_summary <- data.frame(
  Variable = c("Exercise minutes", "Brain fog"),
  Type = "Quantitative",
  Category = "",
  N = as.character(sapply(numeric_vars, function(x) sum(!is.na(x)))),
  Percent = "",
  Mean = as.character(round(sapply(numeric_vars, function(x) mean(x, na.rm = TRUE)), 2)),
  SD = as.character(round(sapply(numeric_vars, function(x) sd(x, na.rm = TRUE)), 2)),
  Median = as.character(round(sapply(numeric_vars, function(x) median(x, na.rm = TRUE)), 2)),
  Min = as.character(round(sapply(numeric_vars, function(x) min(x, na.rm = TRUE)), 2)),
  Max = as.character(round(sapply(numeric_vars, function(x) max(x, na.rm = TRUE)), 2))
)

categorical_table <- function(data, variable, label) {
  data %>%
    filter(!is.na(.data[[variable]])) %>%
    count(.data[[variable]], name = "N") %>%
    mutate(
      Variable = label,
      Type = "Categorical",
      Category = as.character(.data[[variable]]),
      Percent = paste0(round(100 * N / sum(N), 1), "%"),
      Mean = "",
      SD = "",
      Median = "",
      Min = "",
      Max = "",
      N = as.character(N)
    ) %>%
    select(Variable, Type, Category, N, Percent, Mean, SD, Median, Min, Max)
}

categorical_summary <- bind_rows(
  categorical_table(survey, "age", "Age group"),
  categorical_table(survey, "gender", "Gender"),
  categorical_table(survey, "academic_program", "Academic program"),
  categorical_table(survey, "exercise_type", "Preferred exercise type"),
  categorical_table(survey, "intensity_level", "Exercise intensity")
)

summary_table <- bind_rows(
  numeric_summary,
  categorical_summary
)

summary_table %>%
  knitr::kable(
    caption = "Descriptive summary statistics for Group 10 variables",
```

```
    booktabs = TRUE,
    align = "lllrrrrrrr"
  ) %>%
  kableExtra::kable_styling(
    latex_options = c("striped", "hold_position", "scale_down"),
    full_width = FALSE
  )
```

```
## Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be
## resized.
```

Table 1: Descriptive summary statistics for Group 10 variables

Variable	Type	Category	N	Percent	Mean	SD	Median	Min	Max
Exercise minutes	Quantitative		209		195.2	195.43	150	0	1100
Brain fog	Quantitative		209		2.98	0.91	3	1	5
Age group	Categorical	18-20	3	1.4%					
Age group	Categorical	21-23	30	14.4%					
Age group	Categorical	23-25	55	26.3%					
Age group	Categorical	25-30	98	46.9%					
Age group	Categorical	30-35	15	7.2%					
Age group	Categorical	over 35	8	3.8%					
Gender	Categorical	Female	53	25.4%					
Gender	Categorical	Male	156	74.6%					
Academic program	Categorical	Business Informatics	32	15.3%					
Academic program	Categorical	Data Science	177	84.7%					
Preferred exercise type	Categorical	Cardio	75	39.1%					
Preferred exercise type	Categorical	Team Sports	46	24%					
Preferred exercise type	Categorical	Weightlifting	49	25.5%					
Preferred exercise type	Categorical	Yoga/Flexibility	22	11.5%					
Exercise intensity	Categorical	Low-intensity	22	11.5%					
Exercise intensity	Categorical	High-intensity	170	88.5%					

The next table summarizes the main categorical variable, preferred type of exercise.

```
exercise_type_summary <- survey %>%
  filter(!is.na(exercise_type)) %>%
  count(exercise_type, name = "n") %>%
  mutate(percent = round(100 * n / sum(n), 1)) %>%
  arrange(desc(n))

exercise_type_summary
```

```
## # A tibble: 4 x 3
##   exercise_type        n percent
##   <fct>            <int>   <dbl>
## 1 Cardio              75    39.1
## 2 Weightlifting       49    25.5
## 3 Team Sports         46    24
## 4 Yoga/Flexibility    22    11.5
```

Analytic inference

Because brain fog is measured on an ordinal 1-5 scale and exercise minutes may not be normally distributed, Spearman's rank correlation is suitable for RQ1 and RQ3. It tests whether two variables have a monotonic relationship. For RQ2, the Wilcoxon rank-sum test (Mann-Whitney U test) is used because it compares the ordinal brain fog variable between two independent intensity-level groups without assuming normality.

```r
test_results <- data.frame(
  Research_question = c("RQ1", "RQ2", "RQ3"),
  Test = c(
    "Spearman correlation",
    "Wilcoxon rank-sum test",
    "Kruskal-Wallis rank-sum test"
  ),
  Hypothesis = c(
    "More exercise minutes are associated with lower brain fog",
    "High-intensity exercise correlates with lower brain fog",
    "Exercise minutes differ by age group"
  ),
  Estimate = c(
    round(unname(rq1_test$estimate), 3),
    NA,
    NA
  ),
  Statistic = c(
    round(unname(rq1_test$statistic), 2),
    round(unname(rq2_test$statistic), 2),
    round(unname(rq3_test$statistic), 2)
  ),
  P_value = c(
    p_text(rq1_test$p.value),
    p_text(rq2_test$p.value),
    p_text(rq3_test$p.value)
  ),
  Decision_5_percent = c(
    ifelse(rq1_test$p.value < 0.05, "Reject H0", "Do not reject H0"),
    ifelse(rq2_test$p.value < 0.05, "Reject H0", "Do not reject H0"),
    ifelse(rq3_test$p.value < 0.05, "Reject H0", "Do not reject H0")
  )
)


test_results %>%
  rename(
    RQ = Research_question,
    `P value` = P_value,
    `Decision 5 percent` = Decision_5_percent

  ) %>%
  mutate(
    Hypothesis = kableExtra::linebreak(Hypothesis, align = "l")
  ) %>%
  knitr::kable(
    caption = "Analytic inference results",
```

```r
    booktabs = TRUE,
    escape = FALSE,
    align = "lllrrll"
  ) %>%
  kableExtra::kable_styling(
    latex_options = c("striped", "hold_position"),
    full_width = FALSE
  ) %>%
  kableExtra::column_spec(1, width = "1cm") %>%
  kableExtra::column_spec(2, width = "3cm") %>%
  kableExtra::column_spec(3, width = "3cm")
```

Table 2: Analytic inference results

RQ	Test	Hypothesis	Estimate	Statistic	P value	Decision 5 percent
RQ1	Spearman correlation	More exercise minutes are associated with lower brain fog	-0.113	1694191.23	0.051	Do not reject H0
RQ2	Wilcoxon rank-sum test	High-intensity exercise correlates with lower brain fog	NA	1980.00	0.634	Do not reject H0
RQ3	Kruskal-Wallis rank-sum test	Exercise minutes differ by age group	NA	6.31	0.278	Do not reject H0

Interpretation of analytic results

RQ1 result To address our first research question, we performed a Spearman rank correlation test to evaluate the relationship between weekly physical exercise and reported brain fog levels during lectures. This non-parametric test is given that our independent variable is an ordinal scale from 1 to 5, which violates the normality assumptions required for standard linear models.

The test yielded a Spearman correlation coefficient estimate of -0.113, indicating a very weak negative relationship between the two variables. The resulting p-value was calculated to be 0.051. Because this p-value is slightly above our strict 5% significance threshold ($\alpha = 0.05$), we do not reject the null hypothesis (H_0), which states that there is no monotonic association between exercise minutes and brain fog.

The statistical test confirms that we cannot definitively claim a general trend of lower brain fog for students who exercise more across the entire dataset. While our visualization hinted at a protective threshold at the extreme ends, specifically that severe brain fog is isolated to lower exercise levels, the weak correlation coefficient and the borderline non significant p-value demonstrate that this pattern is not strong or consistent enough across the central categories to be statistically reliable.

Ultimately, these findings suggest that while high levels of physical activity may limit severe cognitive fatigue for some, minor fluctuations in weekly exercise minutes do not systematically predict lower levels of day to day brain fog for the broader student body.

RQ2 result To evaluate whether high-intensity training correlates with higher academic confidence (measured by lower brain fog) compared to low-intensity training, we conducted a Wilcoxon rank-sum test (Mann-Whitney U test). This non-parametric test is appropriate because it compares the distribution of an ordinal variable (brain fog on a 1-5 scale) across two independent groups (High-intensity vs. Low-intensity).

The statistical test yielded a p-value of 0.634. Since this p-value is well above the standard 5% significance threshold ($\alpha = 0.05$), we fail to reject the null hypothesis. Statistically, there is no significant shift in the distribution of brain fog scores between students who prefer high-intensity workouts and those who prefer low-intensity workouts.

These analytic results align closely with our visual exploratory findings. While the boxplot showed a slightly more concentrated cluster of responses for the high-intensity group, the median brain fog level was identical (level 3) across both intensity brackets. The Wilcoxon test confirms that the observed minor differences in variance are not statistically meaningful.

Ultimately, we must reject the hypothesis that high-intensity training provides a superior benefit to academic confidence compared to low-intensity training.

RQ3 result The Kruskal-Wallis test has a p-value of r p_text(rq3_test*p.value*).*Atthe*5p.value $<$ 0.05, "reject the null hypothesis and conclude that weekly exercise minutes differ between at least some age groups", "do not reject the null hypothesis, so the data do not provide strong evidence that weekly exercise minutes differ between age groups").

Functions used

Function	Purpose in this report
`read_csv()`	Loads the CSV survey file.
`select()` and `starts_with()`	Extract the demographic and Group 10 variables.
`str_remove()`	Removes the prefixes `DQ` and `G10` from column names.
`rename()`	Gives the selected columns shorter names.
`mutate()`	Converts variables and creates analysis-ready values.
`filter()`	Removes missing values before plots, summaries, and tests.
`ggplot()`	Creates all visualizations.
`geom_boxplot()`, `geom_jitter()`, `geom_point()`, `geom_smooth()`	Show distributions, individual observations, scatterplots, and trend lines.
`count()` and `summarise()`	Compute categorical frequencies and summary statistics.
`cor.test()`	Performs Spearman correlation tests for RQ1 and RQ3.
`kruskal.test()`	Tests whether brain fog differs across exercise-type groups for RQ2.
`knitr::kable()`	Displays results in readable tables.

Conclusion

This report explored the relationship between physical exercise habits and mental brain fog among students. The descriptive analysis summarizes how much students exercise, which exercise types they prefer, and how strongly they report brain fog during lectures. The exploratory plots show the observed patterns visually, while the analytic tests evaluate whether these patterns are statistically supported.

Overall, the findings should be interpreted as associations rather than causal effects. The survey is observational, so it cannot prove that exercise directly causes changes in brain fog. However, the results provide useful evidence about whether students with different exercise habits also report different levels of focus-related difficulty during lectures.